

Calculus and Statistics Tutorial -Unit_3

Questions on DD

Q. 1 Find D.D OF $f(x, y, z) = 4xz^3 - 3x^2yz^2$ at point $(2, -1, 2)$ along Z axis

Q. 2 Find D.D OF $f(x, y, z) = xy + yz + xz$ at point $(1, 2, 0)$

in the direction of vector $i + 2j + 3k$

Q. 3 Find D.D OF $f(x, y, z) = x^2 - y^2 + 2z^2$ at point P $(1, 2, 3)$

in the direction of line PQ where $Q(5, 0, 4)$

Q. 4 Find D.D OF $f(x, y, z) = xy^2 + yz^3$ at point $(2, -1, 1)$ along the line $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-1}{1}$

Questions on Gradient

Q. 1 Find ∇f for the following functions at specified point

a) $f(x, y, z) = 4xz^3 - 3x^4yz^3$ at point $(1, -2, 2)$

b) $f(x, y, z) = 4e^{xy} - 3x^4 \sin(yz^3)$ at point $(1, 2, 3)$

c) $f(x, y, z) = 4e^{x^4y} - 3x^4 \cos(yz^3)$ at point $(-1, 2, -1)$

d) $f(x, y, z) = \text{Log}(4x^3 + y^3 + z^3 - 3xyz)$ at point $(1, 2, 3)$

Questions on Curl & Divergence

Find $\nabla \times f$ & $\nabla \cdot f$ for the following functions at specified point

a) $F^{\rightarrow} = 4xz^3i - 3x^4yz^3j + 4e^{x^4y}k$ at point $(0, -2, 2)$

b) $F^{\rightarrow} = 3x^4 \sin \sin(yz^3)i + yj + 4e^{z^4y}k$ at point $(1, 0, 2)$

c) $F^{\rightarrow} = x^{yz}i + y^{zx}j + z^{yx}k$ at point $(0, -2, 2)$

b) $F^{\rightarrow} = 3x^4 \sin \sin(yz^3)i + yj + 4e^{z^4y}k$ at point $(1, 0, 2)$

Questions on Solenoidal, Rotational Irrotational

Which of the following field are Solenoidal, Rotational Irrotational at specified point

a) $F^{\rightarrow} = 4xz^3i - 3x^4yz^3j + 4e^{x^4y}k$ at point $(0, -2, 2)$

b) $F^{\rightarrow} = 3x^4 \sin \sin(yz^3)i + yj + 4e^{z^4y}k$ at point $(1, 0, 2)$

c) $F^{\rightarrow} = x^{yz}i + y^{zx}j + z^{yx}k$ at point $(0, -2, 2)$

b) $F^{\rightarrow} = 3x^4 \sin \sin (yz^3) i + yj + 4e^{z^4 y} k$ at point $(1,0,2)$.

Questions on Scalar Potential

Q. 1 Prove that $F = (6xy + z^3)i + (3x^2 - z)j + (3xz^2 - y)k$

is irrotational & find scalar potential function ϕ such that $F = \nabla \phi$

Q. 2 Prove that $F = (y^2 \cos x + z^3)i + (2y \sin z - 4)j + (3xz^2)k$

is irrotational & find scalar potential function ϕ such that $F = \nabla \phi$